

Chapter 36: Nonlinear Kinetic Alfvén Solitary Waves in Dusty Plasmas with Charge-Varying Dust Grains and Non-Extensive Electrons

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Abstract

The influence of dust charge fluctuations on the propagation characteristics of arbitrary-amplitude kinetic Alfvén solitary waves (KASWs) is investigated in a magnetized dusty plasma consisting of non-extensive electrons, ions, and charge-varying dust grains. Employing the Sagdeev pseudo-potential approach, an energy-balance equation is derived to examine the existence and nonlinear properties of solitary wave structures. The effects of key plasma parameters, including the ratio of the plasma thermal pressure to the magnetic pressure (β), dust concentration parameter (f_2), and non-extensive parameter (q), on the amplitude of the solitary waves are analyzed numerically. The results reveal that the plasma system supports compressive sub-Alfvénic kinetic Alfvén solitons, whose characteristics are significantly modified by dust charge variation and electron non-extensivity. These findings contribute to a better understanding of nonlinear wave phenomena in magnetized dusty plasmas and may be relevant to a variety of astrophysical and space plasma environments.

Keywords. Kinetic Alfvén waves; charge-varying dust; Sagdeev pseudopotential; solitary waves; magnetized plasma; non-extensive electrons.

1. Introduction

Alfvén waves are low-frequency transverse electromagnetic waves that propagate along magnetic field lines and play an important role in ion heating and energy transport in space plasmas. In low-beta plasmas, ($\frac{m_e}{m_i} \ll \beta = \frac{2\mu_0 n_{i0} K T_e}{B_0^2} \ll 1$), these waves become kinetic Alfvén waves (KAWs), which can evolve into nonlinear structures such as solitons, double layers, and shocks. The first exact solution for solitary kinetic Alfvén waves (SKAWs) was obtained by Hasegawa and Mima [1], and since then extensive studies have been carried out in electron-ion plasmas [2–7]. Observations from the Freja satellite suggest that auroral low-frequency turbulence contains electromagnetic spikes with solitary characteristics that may be interpreted as SKAWs [8].

Dusty plasmas, consisting of electrons, ions, and charged dust grains, occur naturally in planetary rings, magnetospheres, and cometary tails. The presence of dust significantly modifies KAW propagation. Several studies have investigated dust-modified KAWs and SKAWs in dusty plasmas [9–11]. However, since dust grains can acquire fluctuating charges, dust charge variation should be treated as an additional dynamical variable. The Orbital Motion Limited (OML) theory is commonly used to determine dust charging. The effects of dust charge fluctuations on Alfvén waves and SKAWs have been examined in both kinetic and fluid descriptions [12–14].

Non-extensive statistics, introduced by Tsallis [15], provide an effective framework for describing plasmas with long-range interactions and memory effects. Such distributions are relevant in many astrophysical environments, including Earth's bow shock, planetary magnetospheres, and the lunar vicinity [16,17]. The influence of non-extensive electrons on SKAWs has been studied in finite-beta plasmas [18], and dust acoustic waves in charge-varying dusty plasmas with non-extensive electrons have also been investigated [19].

In the present work, we study the combined effects of dust charge variation and electron non-extensivity on solitary kinetic Alfvén waves in a low but finite-beta dusty plasma with stationary dust

grains. Using the Sagdeev potential approach, an energy integral is derived and the conditions for the existence of solitary structures are analysed.

2. Basic Set of Governing Equations

We consider a magnetized, collisionless, three component plasma consists of warm positive ions (number density n_i and ion temperature T_i), non-extensive electrons (number density n_e and electron temperature T_e) and massive negatively charged dust particles (number density n_d). The magnetic field is assumed to be directed along z -axis. The plasma β ($= \frac{2\mu_0 n_{i0} K T_e}{B_0^2}$) is considered to be such that $\frac{m_e}{m_i} \ll \beta \ll 1$, where $m_e(m_i)$ is the mass of an electron (ion). Due to very large inertia, the dust grains do not participate in the motion, in the case of studying. We assume all the variations to be in the $x - z$ plane, except the magnetic field is allowed perturbation in the y -direction. In low beta plasma two potential theories can be used incorporating parallel ($\psi_{\parallel}$) and perpendicular ($\phi_{\perp}$) potential functions.

The equations of continuity and motion governing the ion dynamics in this plasma system is described by

$$\frac{\partial n_i}{\partial t} + \frac{\partial(n_i v_{ix})}{\partial x} + \frac{\partial(n_i v_{iz})}{\partial z} = 0 \quad (1)$$

$$\frac{\partial v_{ix}}{\partial t} + v_{ix} \frac{\partial v_{ix}}{\partial x} + v_{iz} \frac{\partial v_{ix}}{\partial z} = -\frac{e}{m_i} \frac{\partial \phi_{\perp}}{\partial x} + \frac{e B_0}{m_i} v_{iy} - \frac{K T_i}{m_i n_i} \frac{\partial n_i}{\partial x} \quad (2)$$

$$\frac{\partial v_{iy}}{\partial t} + v_{ix} \frac{\partial v_{iy}}{\partial x} + v_{iz} \frac{\partial v_{iy}}{\partial z} = -\frac{e B_0}{m_i} v_{ix} \quad (3)$$

$$\frac{\partial v_{iz}}{\partial t} + v_{ix} \frac{\partial v_{iz}}{\partial x} + v_{iz} \frac{\partial v_{iz}}{\partial z} = -\frac{e}{m_i} \frac{\partial \psi_{\parallel}}{\partial z} - \frac{K T_i}{m_i n_i} \frac{\partial n_i}{\partial z}, \quad (4)$$

In presence of strong magnetic field, strongly magnetized electrons can freeze into the magnetic field and moves in the direction of magnetic field. Hence, the equation of continuity of electron is,

$$\frac{\partial n_e}{\partial t} + \frac{\partial(n_e v_{ez})}{\partial z} = 0 \quad (5)$$

The electrons are assumed to be inertia less due to their lighter masses. The expression for non-extensive electron density is given by[19]

$$n_e = n_{e0} \left[1 + (q-1) \frac{e \psi_{\parallel}}{K T_e} \right]^{\frac{3q-1}{2(q-1)}} \quad (6)$$

Here v_{jx} and v_{jz} ($j = i, e$) are the fluid velocity of ion and electron in x and z direction respectively, e is the magnitude of the electron charge, c is the speed of light in vacuum, z_i is ion charge number. Faraday's and Ampere's laws are, respectively,

$$\frac{\partial^2}{\partial x \partial z} (\phi_{\perp} - \psi_{\parallel}) = \frac{\partial B_y}{\partial t} \quad \text{and} \quad \frac{\partial B_y}{\partial x} = \mu_0 J_z \quad (7)$$

$$\text{The current continuity equation is } \frac{\partial J_z}{\partial z} = e \frac{\partial(n_i v_{iz})}{\partial z} - e \frac{\partial(n_e v_{ez})}{\partial z}, \quad (8)$$

where J_z is the parallel ion current density in z direction. The quasi-neutrality condition is given as

$$z_d n_d + n_e = n_i \quad (9)$$

where z_d is the number of charges residing on the dust grain surface.

3. Dust Charging Equation

The dust grains are charged by the plasma currents that originates from the electrons and ions hitting the grain surface. According to the standard orbit limited probe model, the variable dust charge $q_d = -z_d e$ is determined by the formula

$$\frac{\partial q_d}{\partial t} = I_e(q_d) + I_i(q_d) \quad (10)$$

where I_e and I_i are the average macroscopic electron and ion currents entering the grain surface. The ion current from the thermal ions[20] is

$$I_i = \pi r_d^2 e n_{i0} \sqrt{\frac{8KT_i}{\pi m_i}} \left(1 - \frac{e^2 z_d}{r_d K T_i}\right) \exp\left(-\frac{e\psi_{||}}{K T_i}\right) \quad (11)$$

And electron current from non-extensive electrons is given by[19]

$$I_e = -\pi r_d^2 e n_{e0} \sqrt{\frac{8KT_e}{\pi m_e}} B_q \left[1 + (q-1) \left(\frac{e^2 z_d}{r_d K T_e} + \frac{e\psi_{||}}{K T_e}\right)\right]^{\frac{1}{q-1}+2} \quad (12)$$

$$\text{where } B_q = \begin{cases} \frac{(1-q)^{3/2}}{q(2q-1)} \frac{\Gamma\left(\frac{1}{1-q}\right)}{\Gamma\left(\frac{1}{1-q}-\frac{3}{2}\right)} & \text{for } -1 < q < 1 \\ \frac{(3q-1)}{2} \frac{(q-1)^{3/2}}{q(2q-1)} \frac{\Gamma\left(\frac{1}{q-1}+\frac{3}{2}\right)}{\Gamma\left(\frac{1}{q-1}\right)} & \text{for } q > 1 \end{cases}$$

Here, r_d is the dust grain radius and Γ is the gamma function. It should be noted that we have assumed that the dust grain radius is very less than the electron gyro-radius, so that the charging characteristic is not significantly influenced by existence of external magnetic field and the curvature effect of the trajectory of an electron/ion impinging on a dust grain can be neglected[21].

In the present study we shall consider the dust grain charging process is far faster (typically of order 10^{-8} second) than the remaining characteristic time of Alfvén formation and propagation[22]. This indicates the dust charge at each point of dust acoustic solitons is fixed and it is determined by the plasma parameters at that point[23]. So, it can be considered

$$I_e(q_d) + I_i(q_d) = 0. \quad (13)$$

4. Normalization

After normalizing densities by corresponding equilibrium densities, velocities by Alfvén velocity $V_A = \frac{B_0}{\sqrt{\mu_0 n_{i0} m_i}}$, potential by $\frac{KT_e}{e}$, time by inverse of cyclotron frequency $\Omega^{-1} = \frac{eB_0}{m_i}$ and distances by gyroradius $V_A \Omega^{-1}$, K being the Boltzmann constant we get the normalized set of governing equations as

$$\frac{\partial N_i}{\partial T} + \frac{\partial(N_i V_{ix})}{\partial X} + \frac{\partial(N_i V_{iz})}{\partial Z} = 0 \quad (14)$$

$$\frac{\partial V_{ix}}{\partial T} + V_{ix} \frac{\partial V_{ix}}{\partial X} + V_{iz} \frac{\partial V_{ix}}{\partial Z} = -\frac{\beta}{2} \left(\frac{\partial \phi}{\partial X} + \frac{\alpha}{N_i} \frac{\partial N_i}{\partial X}\right) + V_{iy} \quad (15)$$

$$\frac{\partial V_{iy}}{\partial T} + V_{ix} \frac{\partial V_{iy}}{\partial X} + V_{iz} \frac{\partial V_{iy}}{\partial Z} = -V_{ix} \quad (16)$$

$$\frac{\partial V_{iz}}{\partial T} + V_{ix} \frac{\partial V_{iz}}{\partial X} + V_{iz} \frac{\partial V_{iz}}{\partial Z} = -\frac{\beta}{2} \left(\frac{\partial \psi}{\partial Z} + \frac{\alpha}{N_i} \frac{\partial N_i}{\partial Z}\right) \quad (17)$$

$$\frac{\partial N_e}{\partial T} + \frac{\partial(N_e V_{ez})}{\partial Z} = 0 \quad (18)$$

$$N_e = [1 + (q-1)\psi]^{\frac{3q-1}{2(q-1)}} \quad (19)$$

After normalization equations (5), (7) and (8) can be combined as

$$\frac{\partial^4(\phi-\psi)}{\partial X^2 \partial Z^2} = \frac{2}{\beta} \left[\frac{\partial^2 N_e}{\partial T^2} + \frac{\partial(N_i V_{iz})}{\partial T \partial Z} \right] \quad (20)$$

Charge neutrality equation (9) becomes

$$f_2 Z_d + N_e f_1 = N_i \quad (21)$$

where $\alpha = \frac{T_i}{T_e}$, $f_1 = \frac{n_{e0}}{n_{i0}}$, $f_2 = \frac{Z_{d0} n_{d0}}{n_{i0}}$ and $\beta = \frac{2\mu_0 n_{i0} K T_e}{B_0^2}$. We have $f_1 + f_2 = 1$

Equation (11-13) can be combined as

$$\sqrt{\alpha Q'} \left(1 - \frac{\psi_d}{\alpha}\right) \exp\left(-\frac{\psi}{\alpha}\right) - f_1 B_q [1 + (q-1)(\psi_d + \psi)]^{\frac{1}{q-1}+2} = 0 \quad (22)$$

where dust surface potential $\psi_d = \frac{e^2 z_d}{r_d K T_e}$ and $Q = \frac{m_e}{m_i}$. The dust surface potential corresponding to unperturbed plasma potential ψ_{d0} can be determined by the equation

$$\sqrt{\alpha Q'} \left(1 - \frac{\psi_{d0}}{\alpha}\right) - f_1 B_q [1 + (q-1)\psi_{d0}]^{\frac{1}{q-1}+2} = 0 \quad (23)$$

Then we have normalized dust charge $Z_d = \frac{\psi_d}{\psi_{d0}}$

Expanding ψ_d near ψ_{d0} by Taylor's series expansion, we can get

$$Z_d = 1 + \gamma_1 \psi + \gamma_2 \psi^2 + \gamma_3 \psi^3 + \dots \quad (24)$$

where $\gamma_1 = \frac{\frac{d\psi_d}{d\psi}}{\psi_{d0}} \Big|_{\psi=0}$, $\gamma_2 = \frac{\frac{d^2\psi_d}{d\psi^2}}{2\psi_{d0}} \Big|_{\psi=0}$, $\gamma_3 = \frac{\frac{d^3\psi_d}{d\psi^3}}{6\psi_{d0}} \Big|_{\psi=0}$

5. Derivation Of Energy Integral

In order to obtain an energy integral, we consider a stationary wave in a moving frame defined by $\xi = x k_x + z k_z - M t$ where $M = \frac{v}{V_A}$ is the Mach Number and $k_x^2 + k_z^2 = 1$, so that $\frac{\partial}{\partial t} \rightarrow -M \frac{d}{d\xi}$, $\frac{\partial}{\partial x} \rightarrow k_x \frac{d}{d\xi}$ and $\frac{\partial}{\partial z} \rightarrow k_z \frac{d}{d\xi}$.

$$(25)$$

Using equation (20), equations (13-17) can be reduced to a set of ordinary differential equations,

$$\frac{d}{d\xi} (-M N_i + k_x N_i V_{ix} + k_z N_i V_{iz}) = 0 \quad (26)$$

$$-\frac{M}{N_i} \frac{dV_{ix}}{d\xi} = -\frac{\beta k_x}{2} \left(\frac{\partial \phi}{\partial \xi} + \frac{\alpha}{N_i} \frac{\partial N_i}{\partial \xi} \right) + V_{iy} \quad (27)$$

$$-\frac{M}{N_i} \frac{dV_{iy}}{d\xi} = -V_{ix} \quad (28)$$

$$-\frac{M}{N_i} \frac{dV_{iz}}{d\xi} = -\frac{\beta k_z}{2} \left(\frac{\partial \psi}{\partial \xi} + \frac{\alpha}{N_i} \frac{\partial N_i}{\partial \xi} \right) \quad (29)$$

And equation (21) becomes

$$\frac{\beta}{2} k_x^2 k_z^2 \frac{d^4(\phi - \psi)}{d\xi^4} = \frac{d^2}{d\xi^2} [f_1 M^2 N_e - M k_z N_i V_{iz}] \quad (30)$$

Integrating with the appropriate boundary condition i.e. $\psi \rightarrow 0$, $\frac{\partial \psi}{\partial \xi} \rightarrow 0$, $V_{iy} = V_{iz} = V_{ix} \rightarrow 0$ as $\xi \rightarrow \infty$ and simplifying all the above equations (26-30), we finally obtain an equation of the form

$$\frac{d}{d\xi} \left[\frac{\beta}{2} \frac{\partial \psi}{\partial \xi} - \left(\frac{M^2}{N_i^3} - \frac{\alpha \beta}{2 N_i} \right) \frac{\partial N_i}{\partial \xi} \right] = \left(1 - \frac{M^2}{k_z^2} \right) \left[(N_i - 1) - \frac{\beta}{2} \frac{k_z^2}{M^2} N_i A(\psi) - \frac{\alpha \beta}{2} \frac{k_z^2}{M^2} N_i (N_i - 1) \right] - \frac{M^2}{k_z^2} (Z_d - 1) f_2 \quad (31)$$

where

$$A(\psi) = \int_0^\psi N_i d\psi = \frac{2f_1}{5q-3} [1 + (q-1)\psi]^{\frac{5q-3}{2(q-1)}} - \frac{2f_1}{5q-3} + f_2 \left(\psi + \frac{\gamma_1}{2} \psi^2 + \frac{\gamma_2}{3} \psi^3 + \frac{\gamma_3}{4} \psi^4 \right)$$

By multiplying equation (31) by $\left[\frac{\beta}{2} \frac{\partial \psi}{\partial \xi} - \left(\frac{M^2}{N_i^3} - \frac{\alpha\beta}{2N_i} \right) \frac{\partial N_i}{\partial \xi} \right]$ and integrating once with appropriate boundary condition we obtain the equation,

$$\frac{1}{2} \left(\frac{\partial \psi}{\partial \xi} \right)^2 + S(\psi) = 0 \quad (32)$$

$$\text{with Sagdeev potential is given by } S(\psi) = \frac{V(\psi)}{[H(\psi)]^2} \quad (33)$$

$$\text{where } H(\psi) = \frac{\beta}{2} - \left(\frac{M^2}{N_i^3} - \frac{\alpha\beta}{2N_i} \right) \frac{\partial N_i}{\partial \psi}$$

$$V(\psi) = - \left(1 - \frac{M^2}{k_z^2} \right) \left[\frac{\beta}{2} (A(\psi) - \psi) - \frac{k_z^2 A(\psi)^2 \beta^2}{8M^2} - \frac{k_z^2 \alpha \beta^2}{4M^2} (B(\psi) - A(\psi)) + M^2 \left(\frac{1}{N_i} - 1 \right) - \right. \\ \left. M221Ni2-1+\alpha\beta2Ni-1-\alpha\beta2\log Ni-kz2\beta2A\psi Ni-\psi-kz2\alpha\beta24M2A\psi Ni-B\psi+\alpha\beta kz22\log Ni-1+1 \right. \\ \left. Ni-kz2\alpha2\beta24M2Ni22-N+12+M2kz2f2\beta2C\psi+M2kz2f2Zd-1M22Ni2+\alpha\beta2\log Ni-D\psi \right]$$

$$B(\psi) = \int_0^\psi N_i^2 d\psi$$

$$C(\psi) = \int_0^\psi (Z_d - 1) d\psi = \left(\frac{\gamma_1}{2} \psi^2 + \frac{\gamma_2}{3} \psi^3 + \frac{\gamma_3}{4} \psi^4 \right)$$

$$D(\psi) = \int_0^\psi \left(\frac{M^2}{2N_i^2} + \frac{\alpha\beta}{2} \log N_i \right) \frac{dZ_d}{d\psi} d\psi$$

Equation (32) is the non-linear equation called “energy integral” and $S(\psi)$ is the Sagdeev potential. The amplitude of the solitary kinetic Alfvén waves is determined by finding the root of the non-linear dispersion relation viz $S(\psi) = 0$ for different values of the parameters $q, \alpha, \beta, f_1, k_z, M$ etc.

6. Condition of Existence of Solitary Kinetic Alfvén Waves

We analyze the condition under which the energy integral leads to a solitary wave solution of kinetic Alfvén waves. For soliton formation, the Sagdeev potential $S(\psi)$ in equation (33) must satisfy the following conditions: $S(\psi) = 0$, $\frac{dS(\psi)}{d\psi} = 0$, $\frac{d^2S(\psi)}{d\psi^2} < 0$ at $\psi = 0$; $S(\psi) = 0$ at $\psi = \psi_m$ and $S(\psi) < 0$ for ψ lying between 0 and ψ_m , where ψ_m is the maximum amplitude of the solitons.

Then if $\psi_m < \psi < 0$, rarefactive solitary waves exists, and if $0 < \psi < \psi_m$ compressive solitary waves exist. From equation (33) it is clear that conditions $S(\psi) = 0$, $\frac{dS(\psi)}{d\psi} = 0$ at $\psi = 0$ are automatically satisfied. The condition $\frac{d^2S(\psi)}{d\psi^2} < 0$ at $\psi = 0$ can be written as

$$\frac{\left(1 - \frac{M^2}{k_z^2} \right) \left[\left(1 - \frac{\alpha\beta k_z^2}{2M^2} \right) \left(\frac{f_1(3q-1)}{2} + f_2\gamma_1 \right) - \frac{\beta k_z^2}{2M^2} \right] - \frac{M^2}{k_z^2} f_2\gamma_1}{\frac{\beta}{2} - \left(M^2 - \frac{\alpha\beta}{2} \right) \left[\frac{f_1(3q-1)}{2} + f_2\gamma_1 \right]} > 0 \quad (34)$$

In order to get solitary wave solution condition (34) must be satisfied.

7. Parametric Analysis

The Sagdeev potential $S(\psi)$ is plotted for different values plasma parameters f_2, β, q, M and k_z ensuring that the existence condition of kinetic Alfvén solitary waves is satisfied. Only compressive, sub-Alfvénic solitons are found in the present plasma model.

f_2 is the ratio of equilibrium number density between dust and ions. Figure. 1 (a) shows the variation of Sagdeev potential $S(\psi)$ against the potential ψ for different values of $f_2 = 0.2, 0.15, 0.1$ taking other parameters fixed. Figure 1 (b) shows the corresponding soliton profile for same parameters as 1(a). From this figure we can observe that, as f_2 increases the amplitude and width of the solitary wave increases. This can be depicted that the increase of negatively charged dust concentration, increases the ratio of number density between ions and electrons and thus, the driving force of solitons provided by ion inertia increases and we got high amplitude solitary waves.

We can observe the effect of dust charge variations in Figure 2(a) and (b). Taking the fixed parameters as $M = 0.1, \beta = 0.08, f_2 = 0.05, k_z = 0.4, \alpha = 0.1$ and $q = 0.9$; it can be seen that when we consider the charge of the dust particles as a variable (blue curve), the amplitude of the soliton becomes higher than the amplitude of soliton when charge of the dust particles is taken constant (red curve), while the depth is narrowed i.e., we find spikier solitons when the charge fluctuation is considered.

The effect of non-extensive parameter q of electrons on the Sagdeev potential is shown in the figure 3(a) and (b) by plotting Sagdeev potential $S(\psi)$ against ψ for different values of q . Figure 3(a) is plotted for $q < 1$ and figure 3(b) is for $q > 1$. From this figure, we can easily notice that as q deviates from $q = 1$ the amplitude of the soliton increases. That is to say, any deviation from the Maxwellian distribution leads to an increase in the amplitude of the solitons. Non-thermal electrons have a higher proportion of particles with high energies compared to a Maxwellian distribution. So, they can create stronger electric fields and potential structures.

Figure 4 shows the characteristics of solitary waves for different values of β , the ratio of kinetic to magnetic pressure. It can be seen from the figure that as β increases amplitude of the soliton decreases i.e., for bigger magnetic field we get bigger waves. To show this we have plotted $S(\psi)$ for $\beta = 0.07, 0.08$ and 0.09 and the other parameters are taken fixed.

Figure 5 indicates that the soliton amplitude decreases with increasing (k_z), i.e., as the propagation direction becomes more aligned with the external magnetic field.

Finally, Figure 6 shows that increasing the Mach number (M) significantly increases the soliton amplitude while reducing its width. Thus, faster solitary waves are taller and narrower, whereas slower ones are shorter and wider.

8. Conclusion

The propagation of kinetic Alfvén solitary waves in a dusty plasma containing stationary charge-varying dust particles, warm ions, and non-extensive electrons is investigated. Using a two-fluid model, the governing equations are reduced to the Sagdeev pseudo-potential equation, from which the existence conditions of solitary waves are obtained. The results show that only compressive sub-Alfvénic solitons exist in this plasma. The soliton amplitude increases with increasing density ratio f_2 , plasma parameter β ,

and deviation of the non-extensive parameter q from the Maxwellian value ($q = 1$). These results may help in understanding nonlinear structures in astrophysical dusty plasmas.

Conflict of Interest

The author has no conflicts of interest to declare that are relevant to the content of this article.

APPENDIX

The values of γ_1 , γ_2 and γ_3 are given by

$$\gamma_1 = \frac{B_q f_1 (2q-1) [1 + (q-1) \psi_{d0}]^{\frac{q}{(q-1)}} + \left(1 - \frac{\psi_{d0}}{\alpha}\right) \sqrt{\frac{Q}{\alpha}}}{-B_q f_1 (2q-1) \psi_{d0} [1 + (q-1) \psi_{d0}]^{\frac{q}{(q-1)}} - \psi_{d0} \sqrt{\frac{Q}{\alpha}}}$$

$$\gamma_2 = \frac{\frac{1}{2} B_q f_1 (2q-1) q (1 + \gamma_1 \psi_{d0})^2 [1 + (q-1) \psi_{d0}]^{\frac{1}{(q-1)}} - \frac{2\gamma_1 \psi_{d0}}{\alpha} \sqrt{\frac{Q}{\alpha}} \left(1 - \frac{\psi_{d0}}{\alpha}\right) \frac{1}{\alpha} \sqrt{\frac{Q}{\alpha}}}{-B_q f_1 (2q-1) \psi_{d0} [1 + (q-1) \psi_{d0}]^{\frac{q}{(q-1)}} - \psi_{d0} \sqrt{\frac{Q}{\alpha}}}$$

$$\gamma_3 = \frac{1}{6 \left[-B_q f_1 (2q-1) \psi_{d0} [1 + (q-1) \psi_{d0}]^{\frac{q}{(q-1)}} - \psi_{d0} \sqrt{\frac{Q}{\alpha}} \right]} \left[B_q f_1 (2q-1) q (1 + \gamma_1 \psi_{d0})^3 [1 + (q-1) \psi_{d0}]^{\frac{2-q}{(q-1)}} + \right.$$

$$\left. 6\gamma_2 B_q f_1 (2q-1) q (1 + \gamma_1 \psi_{d0}) \psi_{d0} [1 + (q-1) \psi_{d0}]^{\frac{1}{(q-1)}} + \frac{3\gamma_1 \psi_{d0}}{\alpha^2} \sqrt{\frac{Q}{\alpha}} + \left(1 - \frac{\psi_{d0}}{\alpha}\right) \frac{1}{\alpha^2} \sqrt{\frac{Q}{\alpha}} - \frac{6\gamma_2 \psi_{d0}}{\alpha} \sqrt{\frac{Q}{\alpha}} \right],$$

And $B(\psi)$ and $D(\psi)$ are

$$B(\psi) = \int_0^\psi N_i^2 d\psi = f_2^2 \psi + f_2^2 \gamma_1 \psi^2 + \frac{f_2^2 \psi^3}{3} (\gamma_1^2 + 2\gamma_2) + \frac{f_2^2 \gamma_1 \gamma_2 \psi^4}{2} + \frac{f_2^2 \gamma_2^2 \psi^5}{5} + \frac{f_1^2 [1 + (q-1) \psi]^{\frac{4q-2}{(q-1)}} - f_1^2}{2(2q-1)} +$$

$$\frac{4f_1 f_2 [1 + (q-1) \psi]^{\frac{5q-3}{2(q-1)}} [35 - 94q + 63q^2 + \gamma_1 (9q-7) \{-2 + (5q-3)\} \psi]}{(5q-3)(7q-5)(9q-7)} + \frac{8f_1 f_2 [35 - 94q + 63q^2 + \gamma_1 (9q-7)]}{(5q-3)(7q-5)(9q-7)} +$$

$$\frac{\gamma_2 [-4(5q-3) \psi + (15 - 46q + 35q^2) \psi^2]}{(5q-3)(7q-5)(9q-7)},$$

$$D(\psi) = \int_0^\psi \left(\frac{M^2}{2N_i^2} + \frac{\alpha\beta}{2} \log N_i \right) \frac{dZ_d}{d\psi} d\psi$$

$$= \int_0^\psi \left[\frac{M^2}{2} \sum_{k=0}^{\infty} (-1)^k (k+1) \frac{f_2^k (1 + \gamma_1 \psi + \gamma_2 \psi^2 + \gamma_3 \psi^3)^k}{f_1^{k+2}} [1 + (q-1) \psi]^{\frac{(k+2)(3q-1)}{-2(q-1)}} - \frac{\alpha\beta}{2} \log[1 + (q-1) \psi] \right] d\psi$$

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FIGURES

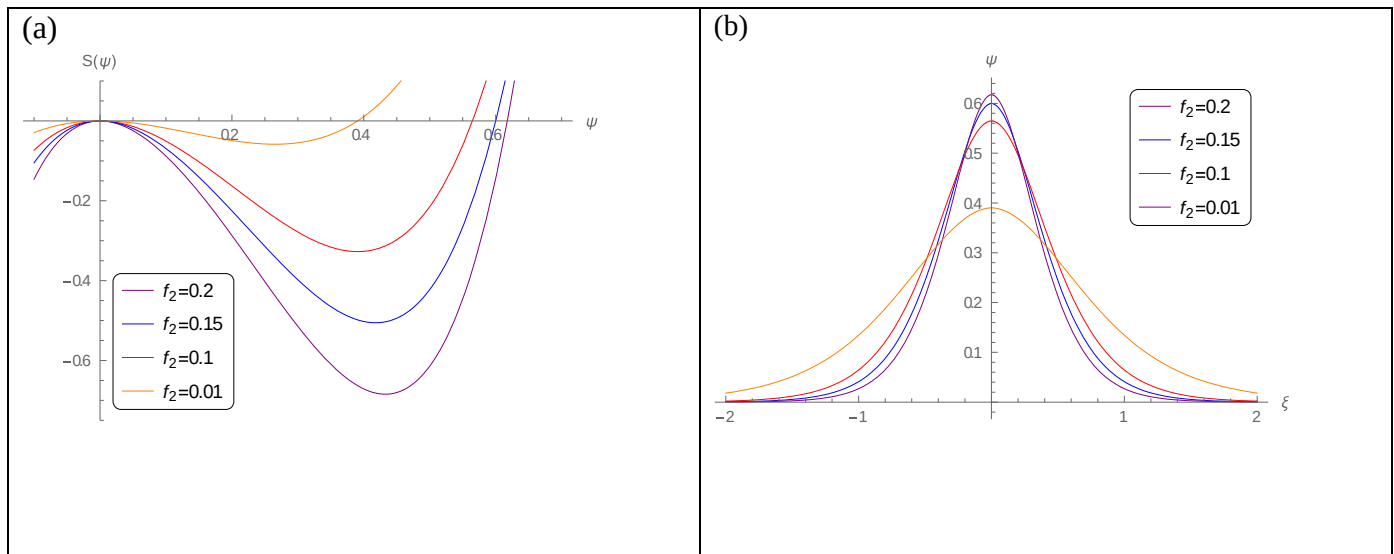


FIG. 1. Plot of (a) the variation of Sagdeev potential $S(\psi)$ against the normalized potential ψ and (b) the normalized potential ψ against ξ for different values of $f_2 = 0.2, 0.15, 0.1, 0.01$ for $q = 0.9$, $M = 0.1$, $k_z = 0.4$, $\beta = 0.08$ and $\alpha = 0.1$.

(a)	(b)
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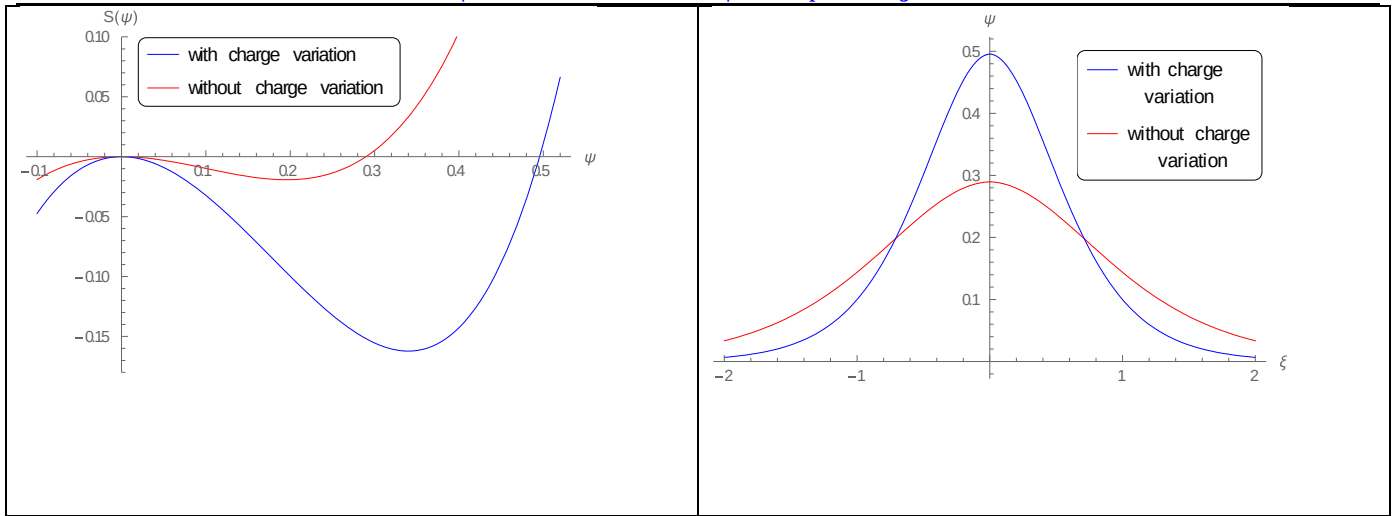


FIG. 2. Plot of (a) the variation of Sagdeev potential $S(\psi)$ against the normalized potential ψ and (b) the normalized potential ψ against ξ considering dust charge variation (blue curve) and considering constant dust charge (red curve) for $f_2 = 0.05$, $q = 0.9$, $M = 0.1$, $k_z = 0.4$, $\beta = 0.08$ and $\alpha = 0.1$

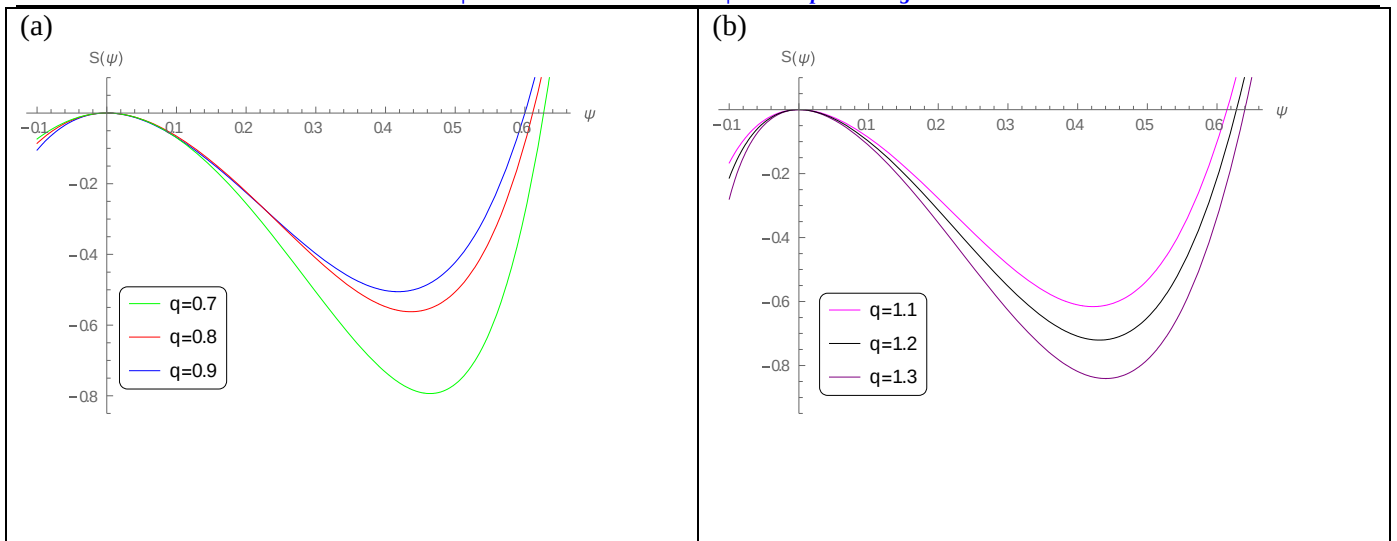


FIG. 3. Plot of the variation of Sagdeev potential $S(\psi)$ against the normalized potential ψ for (a) $q < 1$ (i.e. $q = 0.7, 0.8, 0.9$) and (b) $q > 1$ (i.e. $q = 1.1, 1.2, 1.3$) with $f_2 = 0.15$, $M = 0.1$, $k_z = 0.4$, $\beta = 0.08$ and $\alpha = 0.1$.

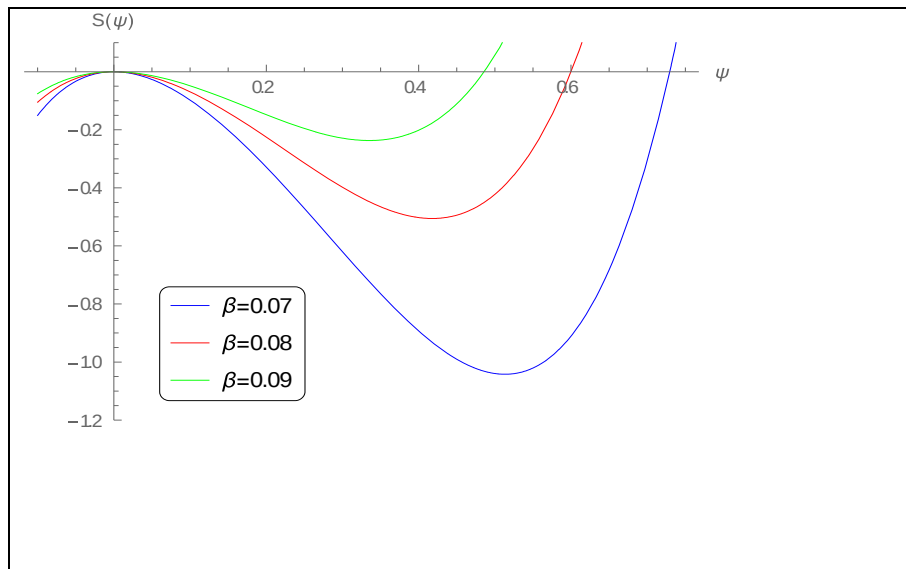


FIG. 4. Plot of the variation of Sagdeev potential $S(\psi)$ against the normalized potential ψ for different values of $\beta = 0.07, 0.08, 0.09$ for $f_2 = 0.15$, $q = 0.9$, $M = 0.1$, $k_z = 0.4$ and $\alpha = 0.1$.

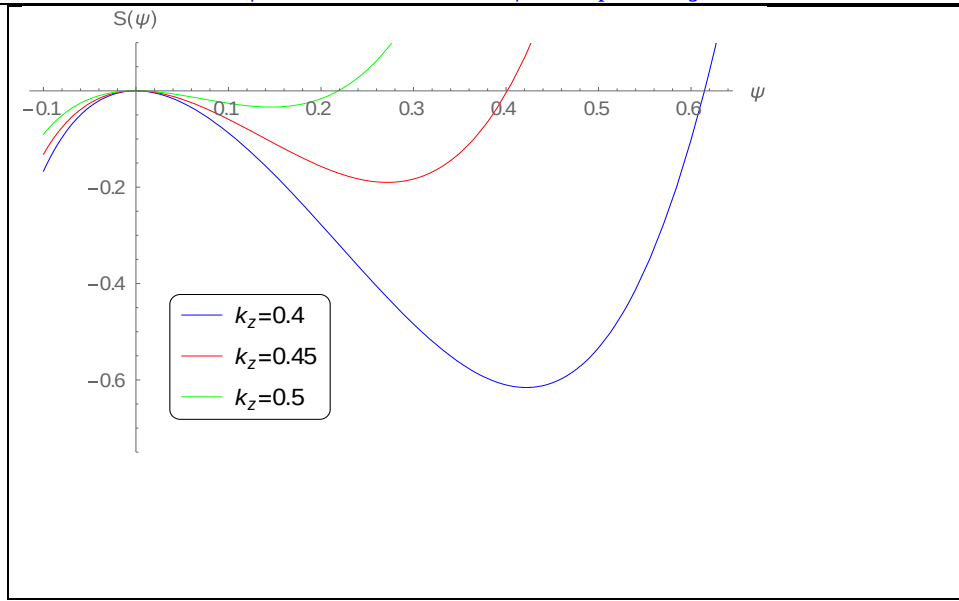


FIG. 5. Plot of the variation of Sagdeev potential $S(\psi)$ against the normalized potential ψ for different values of $k_z = 0.4, 0.45, 0.5$ for $f_2 = 0.15$, $q = 1.1$, $\beta = 0.08$, $M = 0.1$, and $\alpha = 0.1$.

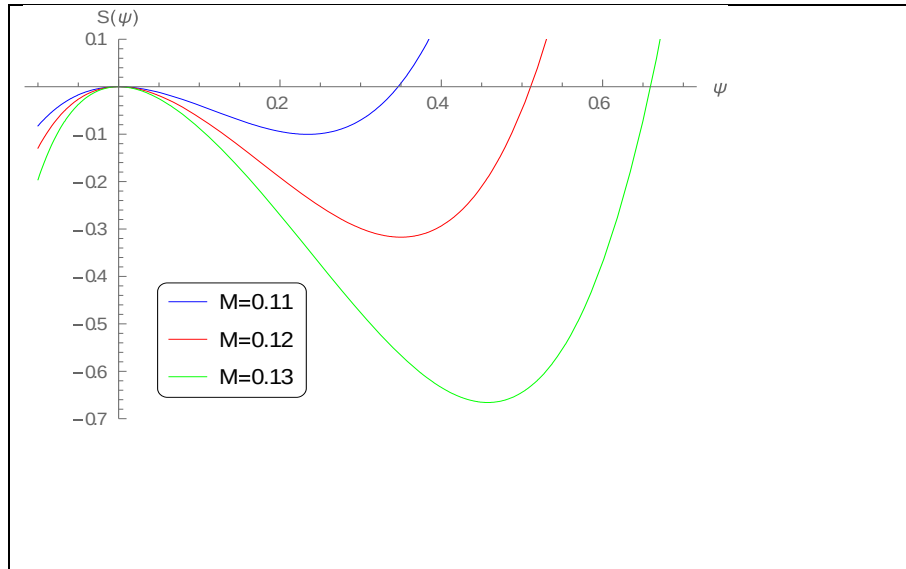


FIG. 6. Plot of the variation of Sagdeev potential $S(\psi)$ against the normalized potential ψ for different values of $M = 0.11, 0.12, 0.13$ for $f_2 = 0.15$, $q = 0.9$, $\beta = 0.08$, $k_z = 0.4$ and $\alpha = 0.1$.